Ex. No. : 04 (a)	Architecture of IoT
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Aim

To study and sketch the architecture of the Internet of Things (IoT) and understand the functions of each layer.

Architecture of IoT

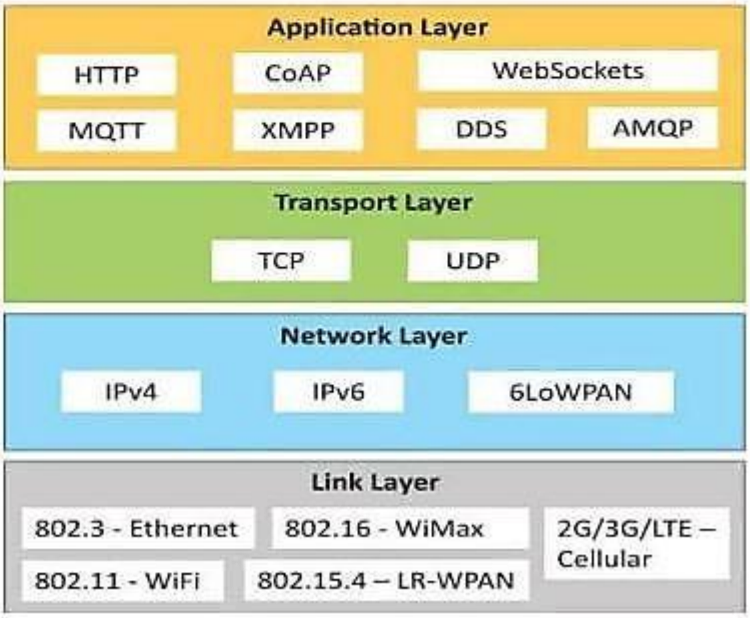

Fig. 1: Architecture of IoT

Description of Layers:

1. Application Layer

   The topmost layer that provides end-user services and IoT applications. It interprets the processed data and delivers meaningful insights through dashboards or control interfaces.
   Examples: Smart home control apps, healthcare monitoring, industrial automation dashboards.

2. Transport Layer

   This layer ensures reliable and efficient communication between IoT devices and cloud servers. It manages end-to-end data transfer using standard internet protocols.
   Common Protocols: TCP (Transmission Control Protocol), UDP (User Datagram Protocol), MQTT, CoAP.

3. Network Layer

   The Network Layer is responsible for routing data packets from source devices to destination servers over the internet. It provides unique addressing using IP/IPv6 and supports low-power wireless networking.
   Technologies: 6LoWPAN, RPL, IP, ICMP.

4. Link (Perception) Layer

   This is the bottom layer, enabling physical connectivity and data sensing. It connects sensors, actuators, and embedded devices to the IoT network. It also handles local communication and medium access control (MAC).
   Examples: Wi-Fi, Bluetooth, Zigbee, RFID, LoRa, NFC.

Result:

Thus, the IoT architecture was demonstrated with seamless data communication and control across different layers in IoT environments.